1. Write a program in Python to accept scores of N players from the user, store them in a list, find and display
   - Minimum score from the list.
   - Maximum score from the list.
   - Average of scores from the list.

   Note: No in-built methods (like min, max, sum) to be used in the program.
2. Write a program in Python to accept names of N students from the user, store them in a list, and arrange the content of the list of names in ascending order. Display the content of the sorted list in two lines (The names starting with alphabets A to M in the first line and the names starting with N to Z in the second line)
3. Write a program in Python to accept names of N cities from the user, store them in a list, and arrange the content of the list of city names in ascending order according to lengths. Display the content of of the sorted list in two lines (The names having length less than 7 characters in the first line and the city name(s) with maximum number of characters in the second line)

   Example:

   INPUTS

```
Number of Cities: 10
Mumbai
Delhi
Bangalore
Hyderabad
Ahmedabad
Chennai
Kolkata
Surat
Pune
Jaipur
```

   OUTPUT

```
Sorted List as per length:
Pune Delhi Surat Mumbai Jaipur Chennai Kolkata Bangalore Hyderabad Ahmedabad

City names with less than 7 characters
Pune Delhi Surat Mumbai Jaipur

City Name(s) with Maximum characters
Bangalore Hyderabad Ahmedabad
```

4. Write a program in Python to accept marks (out of 100) of N students from the user, store them in a list, raise marks by 5% for those which are less than 80.

   Also, find and display
   - Average of marks from modified content of list
   - Count of marks below 50%
   - Count of marks between 70 and 90 (inclusive of 70 and 90)
5. Write a program in Python to accept the number of family members of 5 families from the user, store them in a list, and represent the content of the list as a bar chart using *.

   Example:

```
Enter the number of family members in each of 5 families:
```

5	*****
7	*******
6	******
3	***
4	****

6. Write a program in Python to accept the number of 8 integer values from the user, store them in a list P. Swap each adjacent pair of values in the list P and display the updated content of it.

Sample Example:

```
Enter 8 integer Values:
45 78 34 23 56 87 43 65
Original Content of P:
[45, 78, 34, 23, 56, 87, 43, 65]
Content after swapping of pairs:
[78, 45, 23, 34, 87, 56, 65, 43]
```

7. Write a program in Python
   - To accept names of 6 animals, store them in a list AML & to display the content of the list
   - To re-arrange the content of AML by re-positioning the first half of the list with second half without altering the sequence
   - To display the modified content of AML

```
Pony
Ape
Eagle
Lion
Donkey
Dog
Original content    ->['Pony','Ape','Eagle','Lion','Donkey','Dog']
Re-arranged content ->['Lion','Donkey','Dog','Pony','Ape','Eagle']
```

8. Write a program in Python to accept the number of n number of values from the user, store them in a list VAL. Perform the following operations:
   - In the list VAL, shift the first element to the last and step down each element from 1st element onwards.
   - Display the updated content of VAL
   - Reverse the content of VAL and display the same.
   - In the list VAL, shift the last element to the first and step up each element from 0th element onwards.
   - Display the updated content of VAL

Sample Example:

```
Enter N: 5
45 78 34 23 56
Original Content VAL : [45, 78, 34, 23, 56]
Step Down VAL : [78, 34, 23, 56, 45]
Reversed VAL : [45, 56, 23, 34, 78]
Step Up VAL : [78, 45, 56, 23, 34]
```

IMPORTANT: REFER TO MORE QUESTIONS GIVEN IN THE ASSIGNMENT BOOKLET

General Instructions:

i. Type and execute the solutions of the above mentioned problems on IDLE/colab

ii. Type the following on top of your program code with desired information about each of your programs as comment line (in the same format) - It is mandatory to use **Courier New/Fixed Size Font** with **Style – BOLD & Size – 11** or **12** in all the programs and also **use single line spacing** to avoid wastage of papers:

```
'''Program No        : 01 (Practical List 4)
   Developed By      : Charvi Anand
   Class Section     : XI H
   Date              : 31-Oct-2024'''
```

iii. On successful execution, copy and paste the sample output at the bottom of the program as comment lines. Give program filename as per list and practical number as L4P1.PY, L4P2.PY,...

iv. Save all the programs in a google folder with a name as <Section>-<YourName>-CS (Example: XI-H-RAMESH-CS) and share with CS Teacher. Take a hard copy (printout) of the program and get it signed from the respective CS teacher along with an index entry in a Practical File.

```
# ------------------------------------------------------------#
# List-Program No   : L4-P1
# Developed By      : Puja Dey
# Date              : 19-October-2024
# ------------------------------------------------------------#
L=[]
N=int(input("How many scores?: "))
for i in range(1,N+1,1):
  score=int(input("Enter your scores:"))
  L.append(score)

min_score = L[0]
max_score = L[0]
total_score = 0

for score in L:
    if score < min_score:
        min_score = score
    if score > max_score:
        max_score = score
    total_score += score
average_score = total_score / N

print("Minimum score:", min_score)
print("Maximum score:", max_score)
print("Average score:", average_score)
```

```
How many scores?: 5
Enter your scores:100
Enter your scores:100
Enter your scores:99
Enter your scores:99
Enter your scores:98
Minimum score: 98
Maximum score: 100
Average score: 99.2
```

```
# ------------------------------------------------------------#
# List-Program No   : L4-P2
# Developed By    : Puja Dey
# Date            : 19-October-2024
# ------------------------------------------------------------#
L=[]
N=int(input("How many students: "))
for i in range(1,N+1,1):
  Names=input("Enter name of students: ")
  L.append(Names)
L.sort()
print(L)
print("Names A-M: ")
for j in L:
  if j[0]<="M":
    print(j, end=" ")
print()
print("Names N-Z: ")
for k in L:
  if k[0]>"M'":
    print(k, end=" ")
```

```
How many students: 5
Enter name of students: ROSE
Enter name of students: JASMINE
Enter name of students: BENZENE
Enter name of students: TOLUENE
Enter name of students: METHANOL
['BENZENE', 'JASMINE', 'METHANOL', 'ROSE', 'TOLUENE']
Names A-M:
BENZENE JASMINE METHANOL
Names N-Z:
ROSE TOLUENE
```

```
# ------------------------------------------------------------#
# List-Program No   : L4-P3
# Developed By    : Puja Dey
# Date            : 19-October-2024
# ------------------------------------------------------------#
L=[]
N=int(input("How many cities: "))
for i in range(1,N+1,1):
  Cities=input("Enter name of Cities: ")
  L.append(Cities)
L.sort(key=len)
print("Sorted List as per length: ")
print(L, sep="\t")
print("City names with less than 7 characters: ")
for j in range(0,N):
  if len(L[j])<=7:
    print(L[j], end=" ")
print()
print("City Name(s) with Maximum characters: ")
for k in range(0,N):
  if len(L[k])>=7:
    print(L[k], end=" ")
```

```
How many cities: 10
Enter name of Cities: Mumbai
Enter name of Cities: Delhi
Enter name of Cities: Bangalore
Enter name of Cities: Hyderabad
Enter name of Cities: Ahmedabad
Enter name of Cities: Chennai
Enter name of Cities: Kolkata
Enter name of Cities: Surat
Enter name of Cities: Pune
Enter name of Cities: Jaipur
Sorted List as per length:
['Pune', 'Delhi', 'Surat', 'Mumbai', 'Jaipur', 'Chennai', 'Kolkata', 'Bangalore',
    'Hyderabad', 'Ahmedabad']
City names with less than 7 characters:
Pune Delhi Surat Mumbai Jaipur Chennai Kolkata
City Name(s) with Maximum characters:
Chennai Kolkata Bangalore Hyderabad Ahmedabad
```

```
# ------------------------------------------------------------#
# List-Program No   : L4-P4
# Developed By      : Puja Dey
# Date              : 19-October-2024
# ------------------------------------------------------------#
L=[]
N=int(input("How many MARKS: "))
for j in range(1,N+1,1):
  M=int(input("Enter Marks (Out of 100): "))
  L.append(M)
for i in range(len(L)):
  if L[i]<80:
    L[i]*=1.05
print("Marks List: ", L)
print("Average: ", sum(L)/N)
count=0
for k in L:
  if k<50:
    count+=1
print("Count of marks below 50%: ", count)
Count=0
for m in L:
  if m>=70 and m<=90:
    Count+=1
print("Count of marks between 70 and 90 (inclusive of 70 and 90): ", Count)
```

```
How many MARKS: 4
Enter Marks (Out of 100): 100
Enter Marks (Out of 100): 60
Enter Marks (Out of 100): 80
Enter Marks (Out of 100): 30
Marks List:  [100, 63.0, 80, 31.5]
Average:  68.625
Count of marks below 50%:  1
Count of marks between 70 and 90 (inclusive of 70 and 90):  1
```

```
# ------------------------------------------------------------#
# List-Program No   : L4-P5
# Developed By    : Puja Dey
# Date            : 19-October-2024
# ------------------------------------------------------------#
L=[]
for i in range(1,6,1):
  F=int(input("Enter No. of family members: "))
  L.append(F)
for j in range(0,5,1):
  print( L[j], "*"*L[j],)
```

```
Enter No. of family members: 5
Enter No. of family members: 7
Enter No. of family members: 6
Enter No. of family members: 3
Enter No. of family members: 4
5 *****
7 *******
6 ******
3 ***
4 ****
```

```
# ------------------------------------------------------------#
# List-Program No   : L4-P6
# Developed By    : Puja Dey
# Date            : 19-October-2024
# ------------------------------------------------------------#
L=[]
for i in range(1,9,1):
  I=int(input("Enter an integer):"))
  L.append(I)
N=len(L)
print(L)
for i in range(0,N,2):
  L[i],L[i+1]=L[i+1],L[i]
print(L)
```

```
Enter an integer):45
Enter an integer):78
Enter an integer):34
Enter an integer):23
Enter an integer):56
Enter an integer):43
Enter an integer):65
Enter an integer):87
[45, 78, 34, 23, 56, 43, 65, 87]
[78, 45, 23, 34, 43, 56, 87, 65]
```

```
# ------------------------------------------------------------#
# List-Program No   : L4-P7
# Developed By    : Puja Dey
# Date            : 19-October-2024
# ------------------------------------------------------------#
L=[]
for i in range(1,7,1):
  AML=input("Enter the name of Animal:")
  L.append(AML)
print(L)
N=len(L)
for i in range(N//2):
  L[i],L[N//2+i]=L[N//2+i],L[i]
print(L)
```

```
Enter the name of Animal:PONY
Enter the name of Animal:APE
Enter the name of Animal:EAGLE
Enter the name of Animal:LION
Enter the name of Animal:DONKEY
Enter the name of Animal:DOG
['PONY', 'APE', 'EAGLE', 'LION', 'DONKEY', 'DOG']
['LION', 'DONKEY', 'DOG', 'PONY', 'APE', 'EAGLE']
```

```
# ------------------------------------------------------------#
# List-Program No   : L4-P8
# Developed By      : Puja Dey
# Date              : 19-October-2024
# ------------------------------------------------------------#
VAL=[]
N=int(input("Enter Num: "))

for i in range(1,N+1,1):
  V=int(input("Enter Num for list: "))
  VAL.append(V)
print("Original Content: ", VAL)

step_down_VAL = []
for i in range(1, len(VAL)):
    step_down_VAL.append(VAL[i])
step_down_VAL.append(VAL[0])
print("Step Down VAL: ", step_down_VAL)

step_down_VAL.reverse()
print("Reversed: ", step_down_VAL )

step_up_VAL = [step_down_VAL[-1]]
for i in range(len(step_down_VAL)-1):
    step_up_VAL.append(step_down_VAL[i])
print("Step Up VAL: ", step_up_VAL)
```

```
Enter Num: 5
Enter Num for list: 45
Enter Num for list: 78
Enter Num for list: 34
Enter Num for list: 23
Enter Num for list: 56
Original Content:  [45, 78, 34, 23, 56]
Step Down VAL:  [78, 34, 23, 56, 45]
Reversed:  [45, 56, 23, 34, 78]
Step Up VAL:  [78, 45, 56, 23, 34]
```